

Neurological Diseases

P.A. Kager, P.J. de Vries
Revision January 2015 by M.C. Brouwer

1	Introduction.....	2
2	Diagnostic considerations	2
3	Problem oriented approach to impaired consciousness	3
4	Lateralisation.....	4
5	Meningitis	5
5.1	Some definitions regarding meningitis	5
5.2	Causes of meningitis:.....	5
5.3	Approach to diagnosing meningitis	6
5.4	Bacterial meningitis	6
5.4.1	Clinical aspects	6
5.4.2	Treatment of bacterial meningitis	6
5.4.3	Prevention of bacterial meningitis	7
5.5	Fungal meningitis.....	7
5.6	Viral meningoencephalitis	7
5.7	Aseptic meningitis,	8
6	Other infections of the nervous system.....	8
7	Other acute conditions with confusion and fever.	9
8	Cases	11

1 Introduction

Neurological syndromes which occur during military deployment usually require an approach which combines aspects of emergency medicine and adequate referral to facilities for hospitalisation and intensive care. In this lecture we will cover the important syndromes. Impaired consciousness as a problem will be used to cover diseases of the central nervous system including meningitis, epilepsy and others. Peripheral nervous system disorders will be discussed in the context of acute conditions which may affect a relatively healthy military population deployed in non-industrialised regions. Simple algorithms to diagnosis and treatment will be offered.

2 Diagnostic considerations

Neurological diseases and disease manifestations occur frequently in tropical areas. Neurological diseases seen in the industrialised world do also occur in the tropics but with different frequencies whereas manifestations and diseases related to poverty and poor hygiene are much more frequent in the developing world.

Thus tropical neurology is about manifestations of viral, bacterial, parasitic and fungal infections but also of malnutrition, trauma and perinatal injury.

Of many conditions we do not accurately know the epidemiology, the magnitude, the influence on daily life, let alone the causes and the ways of prevention. For example, epilepsy is a very important disorder in tropical countries, but in most areas we do not know how important. In some areas in Latin America and Africa prevalence is 10 x the prevalence in industrialised countries. There are many causes for epilepsy: cysticercosis (one of the most common causes of epilepsy world wide), schistosomiasis, paragonimiasis, sparganosis, cerebral malaria and cerebral amoebiasis, tuberculosis, pyogenic, viral, fungal infections, trypanosomiasis, perinatal brain damage, a.o. Manifestations of epilepsy may be different, in most areas they are not well examined and defined. Most patients will visit traditional healers and priests, few receive regular medical treatment.

Bacterial meningitis, cerebral malaria, tetanus and cerebrovascular accidents, related to hypertension, are important causes of death and disability in the tropics.

For the traveller to tropical areas (tourist, expatriate professionals, volunteer, business man) the risk of diseases is different from the risk of the local population. During a relatively short stay in the tropics the most frequent neurological manifestations are those due to acute infections with fevers, pneumonia, rickettsioses, malaria, typhoid fever, viral encephalitis and possibly heat stroke.

Table 1. Differences in occurrence of neurological diseases between developing and industrialized countries.

<i>More common in developing countries</i>	<i>Less common in developing countries</i>
meningitis pyogenic tb	degenerative diseases m. Alzheimer m. Parkinson multiple sclerosis
Viral encephalitis	
Tetanus (neonatal)	hernia
rabies	manifestations of malignant diseases
poliomyelitis	
neuropathies nutrition toxic	
complications of haemoglobinopathies	
complications of infections -malaria -trypanosomiasis -relapsing fever -rickettsioses -cysticercosis -echinococcosis -schistosomiasis -amoebiasis.	

3 Problem oriented approach to impaired consciousness

In the following the approach to a patient with impaired consciousness will be discussed. That will include the discussion of meningitis. Although not all patients with meningitis will develop unconsciousness this does not affect the principles of the approach.

STEP 1

Impaired consciousness is a medical emergency and should be handled as such: check for ventilation and circulation and then consider if the cause of coma is evident.

STEP 2

Describe the degree of impairment by the best EMV response (Table 2). Often the patient has to be transported. The EMV response can be used for following up the patient, transferring responsibilities etc.

STEP 3

Then, after physical examination, consider whether there is meningitis and/or whether there are signs of lateralisation.

Table 2 Grading of EMV responses in impaired consciousness.

Response	Adults	Children	
	Glasgow	Adelaide	Blantyre
Eyes open	4 spontaneously 3 to speech 2 to pain 1 never	as Glasgow	1 directed 0 not directed
Best verbal response	5 orientated 4 confused 3 inappropriate 2 incomprehensible 1 none	5 aware of place, gives name & age 4 recognisable and relevant words 3 incomprehensible, but complex vocalisation 2 cry/grunt 1 none	2 appropriate cry 1 inappropriate cry/moan 0 none
Best motor response	6 obeys commands 5 localises pain 4 withdrawal 3 flexes to pain 2 extends to pain 1 none	5 4 --- 3 2 1	--- 2 1 --- --- 0

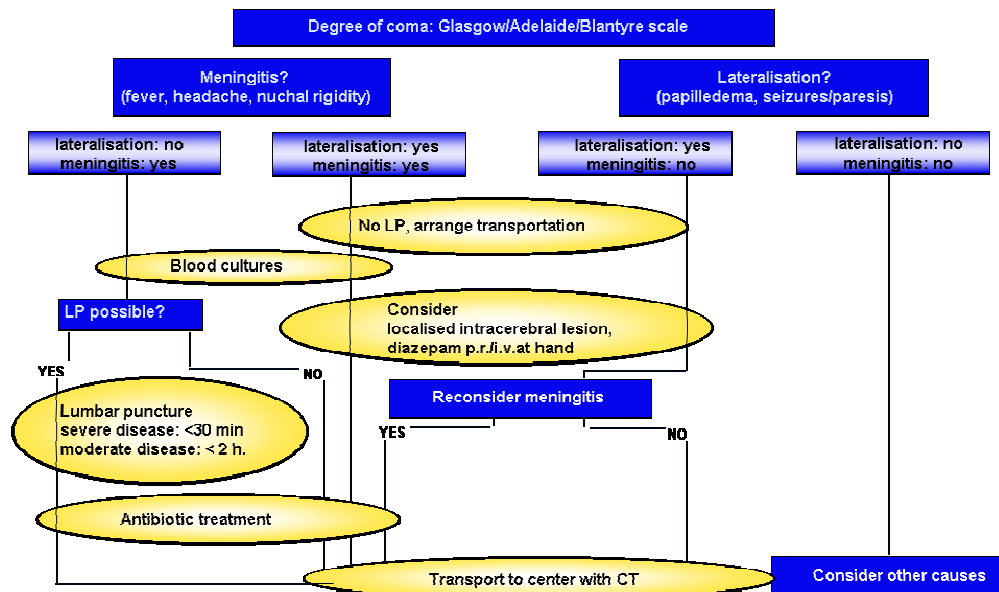


Figure 1 A diagnostic and therapeutic algorithm to coma/impaired consciousness.

4 Lateralisation

Lateralisation is a general term pointing at signs of focal central nervous system disorders such as seizures, hemiparesis etc. These signs are usually found by thorough physical examination. If there is any indication of focal disorders intracerebral mass lesions should be looked for by CT scanning. Intracerebral masses and the consequent raised intracerebral pressure may cause cerebellar herniation after lumbar puncture!

5 Meningitis

5.1 Some definitions regarding meningitis

- movements of the neck in relation to the trunk are ante- and retroflexion, latero-flexion and rotation.
- irritation of the meninges results in reduction of anteflexion.
- neck stiffness, meningeal irritation , a stiff neck with reduction of anteflexion due to an inflammatory process or blood in the subarachnoid space.
- meningitis, infection of the meninges, meningeal irritation with pleocytosis of the CSF (cerebrospinal fluid).
- encephalitis, infection of the cerebrum (encephalos).
- aseptic meningitis, (abacterial meningitis) viral infection of meninges.
- serous meningitis, meningeal irritation and pleocytosis but no micro-organism; due to disease processes close by or in contact with the meninges.
- meningismus, signs of meningeal irritation but no pleocytosis.

5.2 Causes of meningitis:

meningitis	bacterial viral fungal
serous meningitis	brain infarction intracerebral haemorrhage brain abscess epidural abscess septic sinus thrombosis otitis,mastoiditis,osteomyelitis
meningismus	diseases with high fever, infecions (pneumonia, infection with EBV, campylobacter, salmonella)
meningeal irritation	subarachnoid haemorrhage
myogenic,mechanical irritation	trauma discitis drugreactions (phenothiazines) intoxications (lead).

70 to 80 % of cases of bacterial meningitis are caused by *N. meningitis*, *H. influenzae* and *S. pneumoniae*. In the neonatal period *E. coli* and *S. agalactiae* are the most frequently found, also other gramnegative bacilli. Thereafter, until the fifth year *H. influenzae* is a common cause of meningitis. During the last decade vaccination has almost eradicated *H. influenzae* meningitis in many industrialised countries. Tuberculosis remains an important cause of CNS-disease: during miliary tuberculosis, with meningitis and with tuberculomas. Diagnosis is often difficult. In early tb meningitis and in a patient with tuberculoma the CSF may be normal. A low glucose level in the CSF is helpful but a normal level does not exclude tb.

5.3 Approach to diagnosing meningitis

history

physical examination

blood culture

lumbar puncture (when no lateralisation!)

- opening pressure
- number of cells
- protein
- glucose
- gram (ZN)
 - culture
 - other like: lactate, antigen detection, Elisa/PCR.

Table 3 Cerebrospinal fluid

	Leucocytes x 10 ⁶ /l	Protein g/l	Glucose ratio CSF/blood
Normal	0-3(lymfo)	< 0.45	1/2 - 2/3
Viral meningitis	-500 (lymfo)	-1.0	normal
Bact. mening.	3000->> (poly)	0.5-3.0	< 1/2

Traditionally cells counted per counting chamber volume (3 mm³) so that results are often expressed as “count/3”. In children up to 4 weeks old up to 35 cells 10⁶/L is normal, from 4- 8 weeks 25.10⁶/L. From the 8th week the upper limit of normal is 5.10⁶/L. In acute viral meningitis, in 40% there is initially a predominance of polymorphonuclear cells in the CSF, later lymphocytes.

5.4 Bacterial meningitis

5.4.1 Clinical aspects

Differentiation between viral and bacterial meningitis can not be made on the basis of the clinical picture. Viral meningitis is often less-quickly progressive, milder, the patient looks less sick, but.. one can not be sure without lumbar puncture.

The picture of bacterial meningitis in infants and children is not very specific: they are irritable, lose appetite and become lethargic.

In older children and adults we see: headache, fever, reduced consciousness, neck stiffness (nuchal rigidity).

5.4.2 Treatment of bacterial meningitis

As an initial empirical treatment a third generation cephalosporin is advisable (cefotaxim, ceftriaxon, ceftazidim) or a combination of amoxicillin and chloramphenicol. Depending on results of culture and sensitivity one can reduce the spectrum to e.g. penicillin 200000 U/kg/day i.v. in 6 doses (max. daily dose 20 x 10⁶ U) or amoxicillin 200 mg/kg/day in 4 doses (max. 6 g/d.) For beta-lactamase producing *H. influenzae* cephalosporin treatment can be continued. In uncomplicated meningitis treatment duration for pneumococcal or haemophilus infection is 10 days, for meningococcal infection 7 days. Beneficial effect of dexamethasone is proven in high income countries.

Table 4 Most likely pathogen and empirical treatment of bacterial meningitis by age group

Age group	Likely pathogen	Antibiotic treatment
< 3 months	<i>S. agalactiae</i> , <i>E coli</i> , <i>L. monocytogenes</i>	Ampicillin + Cephalosporin
3 months-18 years	<i>N. meningitidis</i> , <i>S. pneumoniae</i> <i>H. influenza</i> (depending on vacc. status)	Cephalosporin (Ampicillin + Camp)
18-50 years	<i>S. pneumoniae</i> , <i>N. meningitidis</i>	Cephalosporin (Ampicillin)
> 50 years	<i>S. pneumoniae</i> , <i>L. monocytogenes</i> Gram-negative bacilli	Ampicillin + Cephalosporin
Impaired cell. immunity	<i>L. monocytogenes</i> , Gram-negative bacilli	Ampicillin + Cephalosporin
Trauma, surgery etc.	<i>Staphylococci</i> , <i>S. pneumoniae</i> Gram-negative bacilli	Vancomycin + Cephalosporin

Recall that many *S. pneumoniae* strains are becoming more and more penicilline and cephalosporin resistant. Therefore vancomycin is often added to the empirical treatment.

5.4.3 Prevention of bacterial meningitis

For household contacts of a patient with meningococcal sepsis or- meningitis, the risk of acquiring meningococcal disease is 500-800 higher than for the population at large. In day-care contacts the risk is also higher (less than in the household). Also about *H.influenzae* meningitis some data exist about increased risk in contacts.

In the Netherlands there is no consensus on chemoprophylaxis. Contacts can be "treated" with rifampicin 2 x 10 mg/kg/day, max. 600 mg/ per dose, infants younger than 1 month 5 mg/kg., during 2 days; for haemophilus rifampicin 20 mg/kg, max. 600 mg, once per day for 4 days. The patient should receive this treatment on leaving the hospital. Treatment of penicillin for meningitis does not clear the throat of bacteria. Vaccination is not discussed in this context.

5.5 Fungal meningitis

The occurrence of fungal meningitis is associated with underlying disease. The most common condition nowadays is AIDS in which cryptococcal meningitis is a common opportunistic infection.

5.6 Viral meningoencephalitis

Viral meningitis may occur in epidemics, caused by enteroviruses, echovirus 9 (with a rash), poliovirus, coxsackie B-virus (muscle pains) or sporadic, e.g. HSV. Vector born "arboviral" (Arthropod Born=ARBO) encephalitis (St Louis, eastern and western equine encephalitis, Japanese B-encephalitis, Früh-sommer meningoencephalitis and other arboviruses) are often epidemic and limited to particular seasons or ecologic conditions. JEB is en/epidemic in many parts of South-east Asia. A vaccine is available. Dengue may cause an encephalitis picture, Lassa fever may have neurological

involvement in the acute stage or in convalescence, with delirium, coma and convulsions.

Rabies is endemic in many parts of the world.

Acute encephalitis often starts with an aspecific clinical picture with headache, changes of consciousness, delirium, coma, convulsions and the classical signs of meningitis may be less clear. CMV encephalitis is a frequent complication of AIDS. Apart from the meningitis viruses malaria and trypanosomiasis may cause a similar disease. This picture is also sometimes seen in trichinosis, cysticercosis, toxocariasis, and during treatment of onchocerciasis and loiasis.

Note:

- viral encephalitis/meningitis:
herpes: acyclovir 30 mg/kg/day i.v. in 3 doses.
- fungal meningitis:
amphotericin B 0.3-0.7 mg/kg/day,

5.7 Aseptic meningitis,

If not viral, also think of partially treated pyogenic infection, tb, mycoplasma, listeria and the late stages of leptospirosis.

6 Other infections of the nervous system.

Very many infectious agents causing diseases of the CNS are found in the tropics. For details on the individual organisms and the respective disease patterns I refer to specific texts. Here we can only give a short overview and some general points of importance. Causes of infection are : viruses, bacteria, protozoa, helminths and fungi.

Viruses: measles, mumps, chickenpox (incidence of measles and subacute sclerosing panencephalitis is decreasing due to vaccination)
HIV and HTLV-I

Bacteria: *T. pallidum*, leptospira, *B. burgdorferi*, *Borrelia recurrentis* and *duttoni*
brucellosis, rickettsiae.

Protozoa: *Plasmodium falciparum*, Trypanosoma spp., *E. histolytica* (and *Naegleria fowleri*)

Helminths: *T. solium*, cysticercosis
E. granulosus, hydatidosis
Gnathostoma spinigerum and *Angiostrongylus cantonensis*
Trichinella spiralis
Schistosoma spp.
Coenurus and *Dracunculus*
Paragonimus westermani
Filariæ

Table 5 Some parasitic infections of the CNS

DISEASE	PARASITE	REMARKS
Amoebic brain abscess	<i>E. Histolytica</i>	brain abscess, rare, often fatal
Primary amoebic meningoencephalitis	Naegleria or Acanthamoeba	meningitis, meningo-encephalitis, by free living amoebae (water)
Trypanosomiasis (African sleeping sickness)	<i>T. gambiense</i> or <i>T. rhodesiense</i>	encephalitis
Trichinosis	<i>T. spiralis</i>	epilepsy; sometimes encephalitis with focal signs
Angiostrongyliasis	<i>Angiostrongylus cantonensis</i>	eosinophilic meningitis
Gnathostomiasis	<i>Gnathostoma spinigerum</i>	lesions in brain and spinal cord; eosinophilic meningitis
Schistosomiasis	<i>S. mansoni</i> <i>S. japonicum</i> <i>S. haematobium</i>	Katayama syndrome; high fever; granuloma (epilepsy); myelitis
Paragonimiasis	<i>Paragonimus westermani</i>	granuloma (epilepsy); not frequent
Cystercercosis	<i>Taenia solium</i>	cysts brain; epilepsy
Echinococcosis	<i>Echinococcus granulosus</i>	cysts brain, spinal cord; not frequent
Coenuriasis	<i>Coenurus cerebralis</i>	space occupying lesion: rare
Dracontiasis	<i>Dracunculus medinensis</i>	extradural tumour; rare

7 Other acute conditions with confusion and fever.

Acute conditions with fever, confusion, changes of consciousness, disorientation, unrest, hallucinations are seen in acute diseases with fever like malaria, pneumonia, typhoid fever a.o. The same symptomatology can be seen at the early stages of meningitis, encephalitis but also in metabolic encephalopathies, in the latter condition without fever. Lumbar puncture will often be required.

note 1: an acute confusional state, also called acute organic brain syndrome, without fever can be caused by intoxications (alcohol, opioids, cocaine a.o.) , hypoglycaemia, shock, liver and kidney failure, vitamin deficiencies, trauma, CVA, subdural haematoma. In acute metabolic and toxic encephalopathies, the CSF is normal.

note 2: severe dehydration can be accompanied with headache, convulsions, coma;

hypoglycaemia is more frequently seen in populations in the tropics than in western people, due to alcohol, malaria, hepatitis and hepatoma, malnutrition;
Reye's syndrome (vomiting, convulsions, coma; is related to intoxication with aflatoxin or use of acetylsalicylic acid (febrile illnesses of children).

note 3: heat stroke, hyperpyrexia due to heat is a severe condition with body temperature above 40 and 41 °C with headache, confusion, aggressive behaviour, delirium, and often death. The mechanism is not well understood; transpiration is reduced. Treatment is: on suspicion cooling with wet sponging and ventilation, bring to cool surroundings, fluids and rehydration. Be aware: there may be an infectious disease, malaria !

Heat exhaustion can develop due to depletion of water, depletion of minerals and lack of sweat production (anhydrotic heat exhaustion). Water depletion and anhydrotic exhaustion can go over into heat stroke.

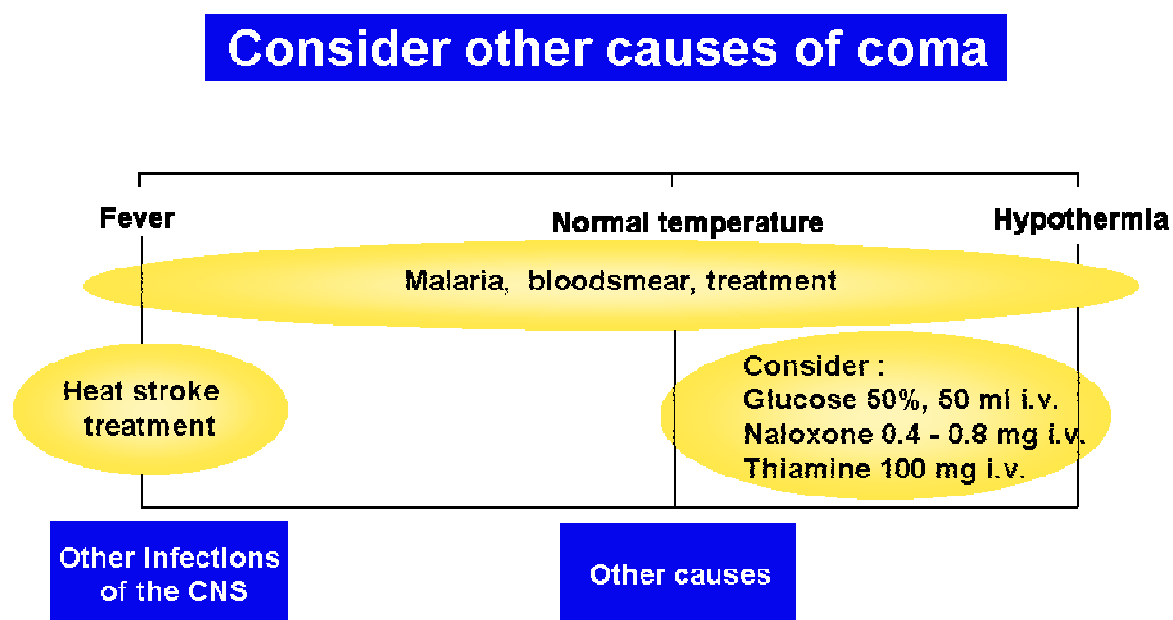


Figure 2 Approach to other causes (non-meningitis) of coma/impaired consciousness.

8 Cases

Briefing for the medical staff.

Field station- 1000 troops of mixed nationality, not adequately prepared.

Keeping peace in 10.000 sq.km. mainly non-combatant, in sub-saharan Africa.

Medical officer- - two nurses- labo technician

First aid-admit some patients

Small labo-a microscope-ECG-ultrasound.

All essential oral drugs + some IV antibiotics and antimalarials.

Referral clinic at 500 km/24 hrs. Emergencies can be evacuated by chopper.

Answer for the following cases the questions:

-What is your most likely diagnosis?

-What is your differential diagnosis?

-What will you do?

Case 1

Male soldier 19 years,

Presentation: coma since 1 hour, didn't wake up

Tent mates: 2 days not feeling well, fever?, malaise, night before vomitus

Phys ex.: 39.5 C, BP 110/70, HR 100/min, Stiff neck, Petechiae conjunctiva, palatum,

Skin: normal, No lymphadenopathy, No focal signs (?)

Fundoscopy: no papilledema

Case 2

Female 36 years, Malaria prophylaxis ++

Fever, occipital headache, neck ache, stiff painful joints

T 39.0 C, BP 120/70

Vague rash, meningeal irritation, no lymphadenopathy, some splenomegaly

Labo: WBC $2.7 \cdot 10^9/L$, 42% lymphocytes, 12% monocytes, platelets $73 \cdot 10^{12}/L$

Case 3

Prostitute, 30 years,

Since 2 weeks slight fever, headache, some visual disturbance

Examination: may be stiff neck, Temp 38C

What is your first idea? What to do?

Case 4

Soldier, on guard during the day,

16.00 aggressive, knocked down a colleague without being hit back, at 16.15 not feeling well, impaired consciousness

Ex: drowsy, hot, red, dry, BP 90/50, HR 120/min, T 41.5 C

No nuchal rigidity

Case 5

Soldier, not waking up, night before party.

Ex.: somnolent, white, moist skin, narrow pupils,

BP 110/80, HR 80/min, T 36.5 C

Case 6

Dutch Caucasian female 27 years, living in southeast Asia

While on leave in the Netherlands: epileptic seizure

ER: phys ex.: confused, weak, but further normal consciousness, some difficulties in talking/finding the correct words, no fever, no nuchal rigidity, nil else of notice

Labo: not remarkable

Case 7

Dutch caucasian male 37 years

Presents with urine retention, faecal gradually progressing to painful perineal region and later irradiating (ischiatric) pain, sensory impairment perineal region, anal sphincter dysfunction and drop foot.

Visited Malawi regularly, last time 4 months before onset of complaints

Phys ex.: healthy, drop foot, hyposensitivity of riding breeches region

Case 8

Dutch female 26 years

2 weeks ago painful peri-anal skin, 1 week ago vesivular lesions,

now: urine retention, anal sphincter dysfunction, painful perineal region, sensory impairment.

Phys ex.: healthy, some residual skin lesions which may have been vesicles, low sphincter tonus, perianal hyposensitivity